Charlie Groll

Toronto, ON · charliegroll04@gmail.com · 647-405-3970
cgroll04.github.io · linkedin.com/in/charlie-groll · github.com/CGroll04

EXPERIENCE

IBM Consulting

Student On Call (part-time) — Sep 2026 - Present

Developer Intern — Sep 2025 - Aug 2026

- Built and integrated a multi-agent intent router in IBM watsonx Orchestrate for a client chatbot, validating and classifying questions and handling ambiguity to produce structured inputs for knowledge-base retrieval.
- Built and deployed a human-in-the-loop, AI-assisted workflow that compares manually supplied sales-dashboard data with previous input and generates formatted change reports used by IBM consultants.
- Prototyped a VS Code chatbot extension with selected-code and file context; tested FastAPI and WebSocket communication approaches.

TD Bank - Data Engineer Intern (Adoption and Enablement) — May 2025 - Aug 2025

- Produced Power BI tutorials for TD employees: researched topics, scripted and recorded narrated demonstrations, incorporated feedback, and published resources to internal SharePoint.

Queen's Engineering Society - Software Developer — Oct 2024 - Apr 2025

- Implemented Firebase Authentication and Cloud Firestore integration for Student-BST, a student-marketplace web app prototype; contributed project research and planning.

Queen's University - Teaching Assistant — Sep 2023 - Apr 2024; Sep 2024 - Apr 2025

- Helped first-year engineering students debug code in introductory programming labs and graded assignments.

PROJECTS

SweetSpots - Co-Founder & Sole iOS Developer · SwiftUI · Firebase · MapKit — Jul 2025 - Present

- Built, released, and maintain a native iOS app for saving and sharing places; reached 508 first-time App Store downloads and 318 registered accounts as of October 2026.
- Implemented Firebase Authentication, real-time Cloud Firestore data, MapKit search, shared collections, deep links, and an iOS share extension.
- Built proximity reminders with Core Location geofencing, permission handling, distance-prioritized monitoring within iOS's 20-region limit, and targeted unit tests.

Spotify Release Tracker - Python · Spotify Web API · Telegram Bot API — Jun 2026

- Built a Python tool using the Spotify Web API and Telegram Bot API to send alerts for new albums, singles, and guest features from playlist artists.
- Implemented OAuth, API pagination, rate-limit handling, release deduplication, and persistent JSON state; documented secure environment-variable setup.

TECHNICAL SKILLS

Languages: Python · Swift

AI & Integration: IBM watsonx Orchestrate · Multi-agent workflows · API integration · OAuth

Application Development: SwiftUI · Firebase · Cloud Firestore · MapKit · Core Location · Git · Xcode

EDUCATION

Queen's University - Bachelor of Applied Science, Computer Engineering — Expected May 2027

Coursework: Machine Learning & Deep Learning (In Progress) · Intro to Data Science · Operating Systems

Awards: Queen's Principal's Scholarship · TD Achieve the Dream Scholarship